PROJECT-BASED LEARNING

PHASE-II EVALUATION

Smart Public Complaint Analyzer

Team ID: DSCPP-III-2026-T031

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T031
Semester: 3rd
Section: ML-1,DS-1,C,ML-2
Domain: *AIML,AIDS,CORE*

Team Members

1. *Devraj Singh Rawat– 2510370241*
2. *Stuti Mahajan– 2510010881*
3. *Archi Panwar– 2510012717*
4. *Rajat– 2510370572*

Mentor

Dr. Prabhdeep Singh **Interactions completed:4** _____

The team is developing a Smart Public Complaint Analyser under the guidance of Dr.Prabhdeep Singh.Devraj is handles the planning,Stuti handles the C/C++ portion,Archi handles the testing and Rajat takes over the developement.

Phase-I Recap & Feedback Incorporated

Phase-I Status

Phase-I mainly focused on problem analysis,requirements and workflow design.

Feedback Received

- *Suggested improving complaint prioritization and management*
- *recommended implementing duplicate complaint detection*
- *encouraged adding complaint statistics and reports.*

Changes Incorporated

- *Improved priority-based complaint handling.*
- *Planned duplicate complaint detection.*
- *Planned complaint statistics and reports to monitor complaint status and resolution progress.*

Problem Statement & Finalized Objectives

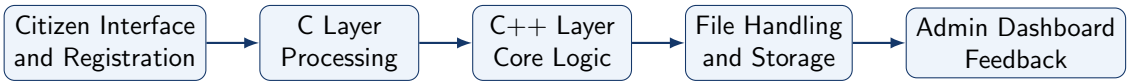
Problem Statement

Public complaint management often faces delays in complaint tracking, prioritization, and department assignment. Our system simplifies complaint registration, tracking, status updates, and complaint management through a centralized platform.

Finalized Objectives

1. *To facilitate the easy raising of complaints and the tracking of their status by the citizens.*
2. *To effectively process, prioritize, sort, search, and route the complaints by using C*
3. *Including Data Structures in C and Object Oriented Programming in C++*
4. *To provide transparency, monitoring by having a dashboard for the admin*

Updated System Architecture / Workflow



Major Components

- *C Engine Core*
- *System Controller/Central Logic*
- *Persistence/UI Layer*

Changes Since Phase-I

- *complain prioritization improved*
- *duolicate detection added*
- *planned statistics and reports*

Module Design & Functional Requirements

Major Modules

1. *Regsitering and Processing complaints*
2. *Detecting Duplicacy*
3. *Assigning Department*
4. *Tracking Status*

Functional Requirements

- *Registering and Categorising*
- *Calculating Priority*
- *Detecting Duplicacy*
- *Storing and Tracking*

Module Dependency

Regeistered complaints are processed and prioritized.Thereafter assigned to departments followed by storing and reporting them.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C / C++*
- Database: *File Handling*
- Framework: *STL*
- Development Tool: *VSCode*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis:Identify complaint registration, categorization, search, routing and tracking requirements
2. System Design: Design C/C++ modules, data structures, workflow and file storage
3. Implementation: Develop C processing modules and C++ OOP, STL and exception handling.
4. Testing: Test registration, search, sorting, routing, file handling and error cases.
5. Evaluation: Evaluate accuracy, response time, reliability and overall effectiveness.

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Arrays, Linked Lists, Trees, Graphs, Hashing, etc.*	*Complaint Storage,Duplicacy,Priority Queue*
OOPs with C++	*Classes, Inheritance, Polymorphism, STL, etc.*	*omplaint processing and status tracking.*

Development & Implementation Progress

Module	Status	Work Completed
Complaint Registration	Completed	*Complaint ID generation*
Complaint Tracking	Completed	*Searching by ID*
Status and Assignment	In Progress	*Improving Status*
Duplicate Detection	Pending	*Duplicate Identification*

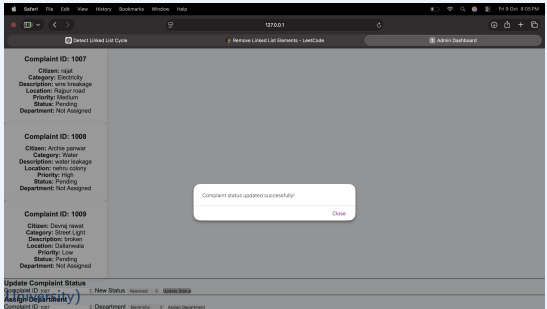
Overall Project Completion: *66%*

Working Prototype / Demonstration

Implemented Features

- *Complaint Registration*
- *Complaint Tracking*
- *Admin Dahsboard*
- *Status Update*

Prototype / Output



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Registering Complaints	*Generation of complaint ID*	*ID generated successfully*	Pass
Complaint Tracking	*Display of IDs*	*Successfully displayed*	Pass
Status Updation	*Updation*	*Success*	*Pass*

Results / Validation

Successful registration of compplaints with traking and status update.Complaint Ids generated and displayed.These tests demonstrate the basic functionality of the complaint management.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: *https://github.com/stutimahajan19-tech/Smart-Public$_{C}$omplaint$_{A}$nalyzer*
- *README.md: Updated*
- *Contributors: 4 Members*
- *Repository Status: Development in progress*
- *Branches: If Applicable*

Development Activity

- *Complaint registration and ID generation*
- *Complaint tracking using Complaint ID*
- *Admin dashboard and status updates*
- *Department and employee assignment*
- *Testing and documentation*

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Devraj Singh Rawat	*Lead*	*Planning and Coordination*	*Project Setup*
Stuti Mahajan	*Developer*	*C/C++ Module Development*	*Data Structures*
Archi Panwar	*DB/Testing*	*Database and Testing*	*Validation & Testing*
Rajat	*Developer,Docs*	*Development and Documentation*	*Website & Documentation*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Data Structure Implementation*
- *Data Synchronisation*
- *Unique complaint IDs*

Solutions / Learning

- *Effiecient data management*
- *Module Integration*

Phase-III Roadmap

1. *Completing pending features*
2. *Integrate all modules to one system*
3. *Testing all features and fixing errors*
4. *End to end system testing*
5. *Demonstration and Documentation*

Expected Outcomes & References

Expected Final Outcomes

- *Complete complaint management prototype*
- *Efficient priority-based complaint handling*
- *Unique complaint ID and status tracking*
- *Improved complaint management and monitoring*
- *Future scope for duplicate detection and analytics*

References

1. *Data Structures and Algorithms — C Programming*
2. *C++ Programming Documentation*
3. *GeeksforGeeks — Linked Lists and Priority Queues*
4. *MDN Web Docs — HTML, CSS and JavaScript*
5. *GitHub — Project Source Code*

Thank You

Questions & Suggestions